Machine Learning

BITS Pilani

Anand Narasimhamurthy

innovate achieve lead

BITS Pilani

ALMLZ565, Machine Learning
Decision trees : An overview

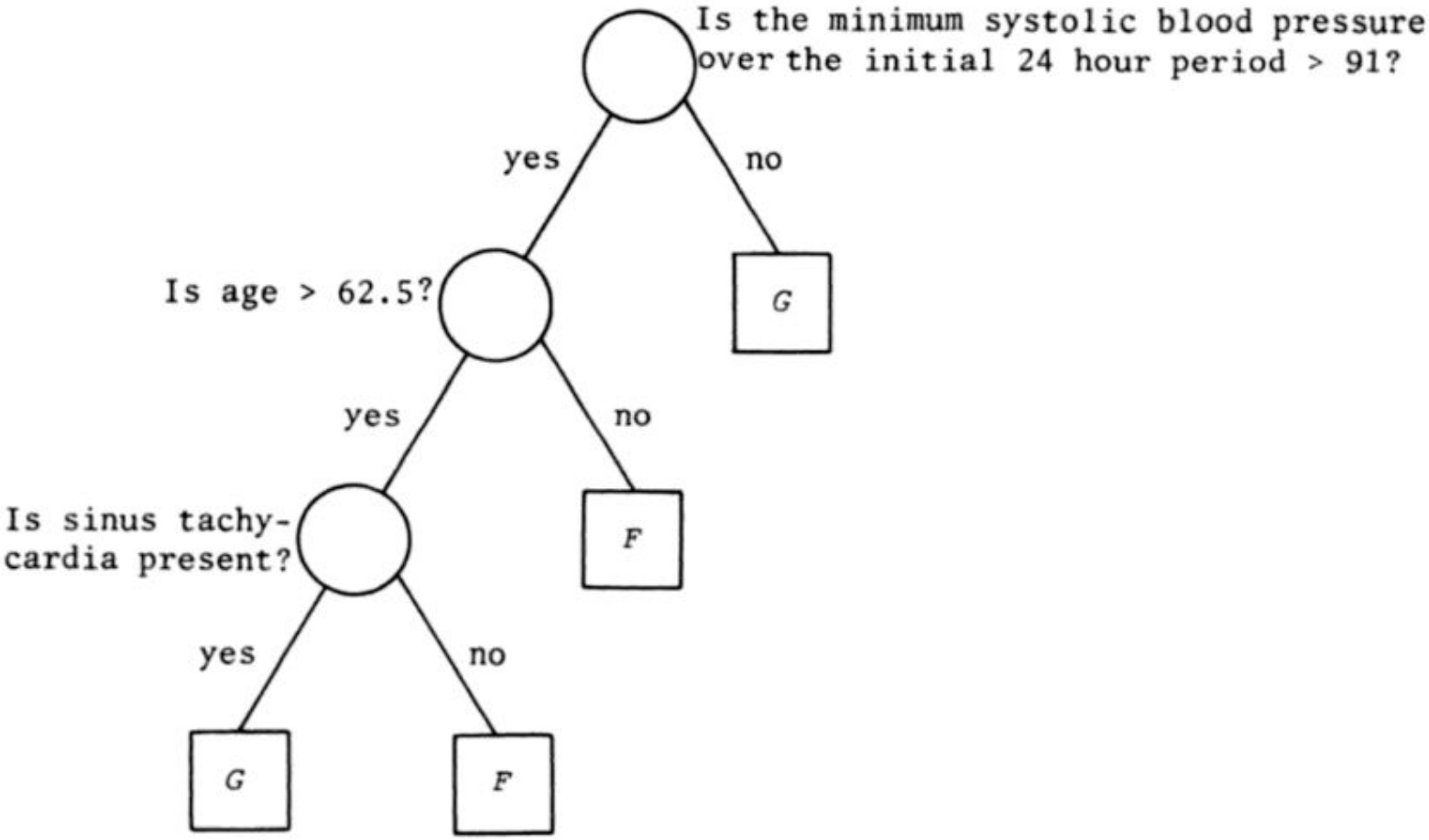
Summary

- **Decision trees can be categorized under non-parametric models.**
- **Primarily a classification model, can be extended to regression problems.**
- **Different tree based models have been proposed, we will focus on Classification and Regression Trees (CART) with binary splits**
- **The decision boundary is computed by successive piecewise axis parallel boundaries**

Practical considerations

- Choice of splits, considerations include impurity measure and criteria to stop growing the tree

An example



A decision tree from seminal paper on Decision tree

Advantages and disadvantages

Advantages

- Can compute complicated nonlinear decision boundaries
- Model is interpretable since the output is typically a set of rules on the attribute values
- Can easily handle a mix of numerical and categorical attributes
- Low computation overhead during classification

Disadvantages

- Prone to overfitting

A few relevant details

Splitting criteria

- **Which attribute should be selected for performing the next split?**
- **Where exactly should the split be performed?**

For a candidate split at a node,

Compute the impurity of that node.

Common choices include entropy and Gini index

Compute the information gain for the candidate split.

The information gain measures how much the impurity decreases if that split is performed.

Select the split (attribute and where to split) which yields the **largest information gain**

Discuss example in https://medium.com/@ompramod9921/decision-trees-6a3c05e9cb82

Common impurity measures

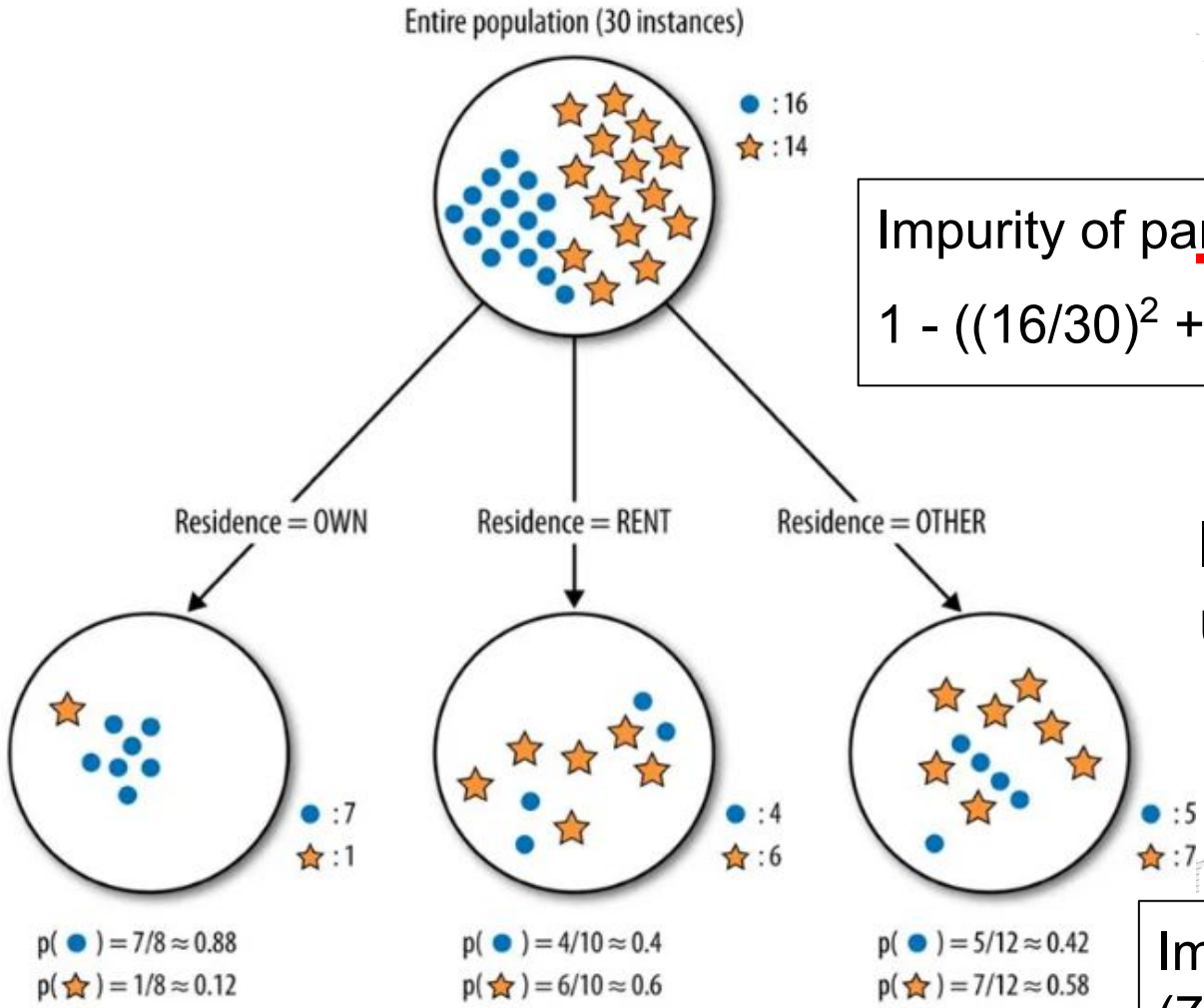
Let p_i denoted the fraction of elements belonging to class i

$$\text{Gini Index} = 1 - \Sigma\, p_i^2$$

$$\text{Entropy} = -\Sigma\, p_i \log_2 p_i$$

Lower value of measure = Lower impurity

i.e. a purer node



Impurity of parent node =

$1 - ((16/30)^2 + (14/30)^2) = 0.4977$

Information gain using Gini index

Impurity = $1 - ((5/12)^2 + (7/12)^2) = 0.4861$

Impurity = $1 - ((7/8)^2 + (1/8)^2) = 0.21875$

Impurity = $1 - ((0.4)^2 + (0.6)^2) = 0.48$

Example from https://medium.com/@ompramod9921/decision-trees-6a3c05e9cb82

Information gain using Gini index

Information Gain = Impurity of parent node - weighted impurities of child nodes

$$= 0.4977 - (8/30 * 0.21875 + 10/30 * 0.48 + 12/30 * 0.4861)$$
$$= 0.4977 - 0.4128$$
$$= 0.0849$$

A few relevant details

Stopping criteria

Typically, node splitting is stopped earlier than achieving zero impurity, as such trees can be long and tend to overfit.

Possible stopping criteria are :

The number of training samples in a node is less than the minimum.

- **Splitting the data set would make children that are below the minimize size.**
- **Splitting criteria improvement is too small.**
- **Tree depth reaches a maximum.**
- **Number of nodes reaches a maximum.**

Variants of decision trees

Most commonly used decision trees are :

CART (Classification and Regression Trees) : Breiman et al., 1984

ID3 : Quinlan, 1983

C4.5 : Quinlan, late 1980s and early 1990s

These differ in the way splits are performed. For instance,

CART only performs binary splits on both numerical as well as categorical values, whereas

C4.5 makes binary splits for numerical attributes and k-way splits for categorical attributes.

References and additional links

- Breiman, L., Friedman, J., Olshen,R. and Stone, C. J. (1984). *Classification and Regression Trees*, Chapman and Hall, New York.

- Quinlan, J. R. C4.5: *Programs for Machine Learning*. Morgan Kaufmann Publishers, 1993.

- Lecture notes on Classification and Regression Trees, University of Toronto

https://www.cs.toronto.edu/~urtasun/courses/CSC411_Fall16/06_trees_handout.pdf

- Documentation on scikit learn

https://scikit-learn.org/stable/modules/tree.html